

Assignment - 6

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Aim : (Dictionaries)

Write a Python program to convert two lists into a dictionary in a way that item from list1 is the key and item from list2 is the value

- b. Write a Python script to generate and print a dictionary that contains a number (between 1 and n) in the form (x, square of x)
- c. Write a Python program to create a dictionary storing movie details such as - (Movie name: show timings) and perform the following operations
 - i. Add a new movie
 - ii. Display movies available at 9 pm
 - iii. Remove the details of the specific movie
 - iv. Removes the last movie details

Theory : In this assignment using `zip()` function we will combine 2 list into a single dictionary and also converting a single dictionary in to 2 lists

Also, using `pop()` and `popitem()` functions we can delete a certain element from the dictionary or the last element from the dictionary respectively.

Program :

```
"""
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2213111"""

#Task-1
#list into dictionary

print("=" * 100)
print("\nTask A\n")
print("\nList into Dictionary\n")
l1 = ["A", "B", "C", "D", "E"]
l2 = [1, 2, 3, 4, 5]

print("1st List is :: ",l1)
print("2nd List is :: ",l2)
res = dict(zip(l1, l2))
```

```

print("The new Dictionary formed using list 1 and 2 is :: ",res)

#dictionary to lists
print("\nDictionary to Lists\n")
x = {1: 'A', 2: 'B', 3: 'C'}
x.items()
lst = list(x.items())
print(lst)
lst1 = list(x.values())
print(lst1)

lst2 = list(x.keys())
print(lst2)

#Task 2
print("=" * 100)
print("\nTask B")
num = int(input("\nEnter Number :: "))
d = dict()
for i in range(1, num+1):
    d[i] = i**2
print("The dictionary with the format as key = number and value = number  
** 2 :: ",d)

#Task 3
print("=" * 100)
print("\nTask C\n")
movie = {'ABCD': '13:00', 'EFGH': '15:00', 'XYZ': '17:00', 'PQRS':  
'21:00'}
print(movie)

# 1. Add
print("\n1. Add")
key = str(input("\nEnter the movie you want to add (Name only):: "))
val = str(input("\nEnter the Time of the movie you want to add (in 24Hr)::  
"))
movie[key] = val
print(movie)

```

```

# 2. Display movies at 9pm
print("\n2. Display movies at 9pm")
dis = str(input('\nThe time at which you want to watch the movie :: '))
value = {i for i in movie if movie[i] == dis}
print(value)

# 3. Remove details of a movie
print("\n3. Remove details of a movie")
rem = str(input('\nEnter the movie you want to delete :: '))
movie.pop(rem)
print(movie)

# 4. Remove last movie
print("\n4. Remove last movie")
print('\nAfter deleting last movie in dictionary :: ')
movie.popitem()
print(movie)

```

Output :

```

=====
Task A

List into Dictionary

1st List is ::  ['A', 'B', 'C', 'D', 'E']
2nd List is ::  [1, 2, 3, 4, 5]
The new Dictionary formed using list 1 and 2 is ::  {'A': 1, 'B': 2, 'C':
3, 'D': 4, 'E': 5}

Dictionary to Lists

[(1, 'A'), (2, 'B'), (3, 'C')]
['A', 'B', 'C']
[1, 2, 3]
=====
Task B

Enter Number :: 4

```

```
The dictionary with the format as key = number and value = number ** 2 ::  
{1: 1, 2: 4, 3: 9, 4: 16}
```

Task C

```
{'ABCD': '13:00', 'EFGH': '15:00', 'XYZ': '17:00', 'PQRS': '21:00'}
```

1. Add

```
Enter the movie you want to add (Name only):: Rudradev
```

```
Enter the Time of the movie you want to add (in 24Hr):: 23:00
```

```
{'ABCD': '13:00', 'EFGH': '15:00', 'XYZ': '17:00', 'PQRS': '21:00',  
'Rudradev': '23:00'}
```

2. Display movies at 9pm

```
The time at which you want to watch the movie :: 21:00
```

```
{'PQRS'}
```

3. Remove details of a movie

```
Enter the movie you want to delete :: Rudradev
```

```
{'ABCD': '13:00', 'EFGH': '15:00', 'XYZ': '17:00', 'PQRS': '21:00'}
```

4. Remove last movie

```
After deleting last movie in dictionary ::
```

```
{'ABCD': '13:00', 'EFGH': '15:00', 'XYZ': '17:00'}
```

Conclusion : In this assignment we learnt how to pop/delete element from a dictionary at a particular index or from the end

Also using zip function to convert 2 lists into dictionary or vice versa